Neuro-Psycho-Contextual Model for Cognitive Aging in Place: From Habit to Spatial Agency

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Scope Statement

Research Highlights & Core Contribution While traditional biomedical paradigms view dementia solely as internal neurobiological deficits, our study proposes an integrative Neuro-Psycho-Contextual (NPC) model. This framework demonstrates how spatial design and environmental scaffolding compensate for damaged episodic memory pathways by activating preserved subcortical procedural memory systems—specifically basal ganglia habit circuits. Synthesizing clinical neuroscience, ecological psychology, and socio-spatial theory, our research reveals how multi-scalar interventions (domestic, neighborhood, urban) transfer processing loads from vulnerable cortical regions to stable environmental cues. Ultimately, we introduce Spatial Cognitive Justice, offering an operational roadmap for policymakers and designers to sustain functional agency, dignity, and independence in aging populations ("Aging in Place"). Alignment with Section Scope Our manuscript aligns with Neurocognitive Aging and Behavior across key dimensions: Neural Foundations of Cognition: Investigates brain-behavior relationships during senescence, exploring how procedural learning mechanisms, sensorimotor loops, and neuroplasticity modulate cognitive trajectories and autonomy. Core Topics Addressed: Connects key domains including learning, memory, cognitive modulation, behavioral interventions, sensorimotor interaction, and social cognition. Methodological Rigor: Moves beyond basic dementia screening tools (MMSE, MoCA) to deliver a nuanced synthesis of subcortical versus cortical mechanics. UN SDGs Support: Advances SDG 3 (Good Health and Well-being) by establishing physical space as an active therapeutic agent.

Conflict of interest statement

The authors declare that the research was conducted in the absence of any commercial or financial relationships that could be construed as a potential conflict of interest

Credit Author Statement

Faeghe Deljoo Ghamgosar: Conceptualization, Formal Analysis, Investigation, Methodology, Software, Visualization, Writing – original draft. **Muhammad Sadra**: Methodology, Supervision, Validation, Writing – review & editing. **Zahra Vahai**: Supervision, Writing – review & editing.

Keywords

cognitive aging, cognitive health, contextual factors, Environmental Scaffolding, neuroplasticity, procedural memory, spatial agency

Abstract

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In review

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The author(s) verify and take full responsibility for the use of generative AI in the preparation of this manuscript. Generative AI was used

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In review

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Abstract

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Keywords

Procedural Memory, Cognitive Aging, Environmental Scaffolding, Neuroplasticity, Cognitive Health, Contextual Factors, Spatial Agency.

Highlights

- NPC model redefines dementia by linking neuroscience with spatial design scales.
- Environmental scaffolding shifts cognitive load to preserved procedural memory.
- Space acts as an active therapeutic agent to sustain agency in dementia.
- Multi-scalar design transforms navigation from conscious choice to habit flow.
- The study introduces Spatial Cognitive Justice as a new policy framework.

Introduction

The human experience is shaped through an inextricable bond with space. Memory, cognition, and identity are not merely intra-cerebral processes but are the products of continuous interaction with lived environments [1]. Space is not a neutral container; rather, it is an active mediator in organizing experience, consolidating habits, and ensuring the continuity of the "self" [2]. Theories of embodied and distributed cognition [2] demonstrate that the body, environment, and mind operate as an integrated system [4], where cognitive functions are stabilized through spatial repetitions and environmental cues [5]. From this perspective, memory is not solely a

mental phenomenon but a spatio-temporal one that persists within the context of place.

Consequently, with the onset of memory disorders and dementia, this fundamental bond between the individual and the environment undergoes a rupture. Dementia, particularly Alzheimer’s disease, does not merely signify a loss of information; it represents a redefinition of the subject's relationship with space, time, and action [6]. In this state, the individual gradually loses not only their past but also their navigation abilities and the continuity of their agentic identity. However, this cognitive decline is not absolute. While episodic memory is severely impaired, procedural memory and deeply rooted habitual behaviors within the body and space maintain their stability [7][8]. This distinction provides a robust foundation for researchers in the management of dementia.

On a philosophical level, this concept is intertwined with the question of identity. By distinguishing between "idem-identity" (sameness) and "ipse-identity" (selfhood), Ricoeur emphasizes that even with the collapse of historical memory, the continuity of identity remains possible through action in the world [9]. In other words, "I am" derives its meaning through the continuity of movement and action in space. Similarly, Bergson, by differentiating "memory-image" from "memory-habit," illustrates that consciousness is sedimented in the body and persists through motor repetitions [10]. These perspectives explain dementia not merely as a "loss" but as a transformation in the mode of being-in-the-world.

Furthermore, emerging evidence in spatial epidemiology suggests that dementia is not solely a neurobiological pathology but a contextualized experience influenced by environmental and social characteristics [11]. Within this framework, the environment evolves from a passive backdrop into a "distributed cognitive scaffold" that affects the brain, safety, and individual behavior [12][13]. According to the theory of neuroplasticity, and in Malabou’s terms, the environment is one of the primary organizers of the brain [14][15]. Therefore, through the stabilization of spatial habits and sensory cues, the environment can contribute to strengthening cognitive reserve and reducing disorientation.

Despite this, dominant biomedical approaches, with their positivistic focus on brain pathology, have failed to improve the quality of life for the elderly [16]. This perspective reduces the human being to a clinical object, focusing on "what has been lost" while ignoring residual capacities and environmental potentials, thereby undermining individual autonomy [15]. As a result, aging and dementia are represented not as part of the life cycle but as biological deviations or defects [6]. This issue gains twofold importance in the context of demographic shifts and the projected elderly population of 2.1 billion by 2050 [12], a transition that imposes a heavy economic burden on health systems and urban planning [17][18]. On the other hand, research on existing environmental interventions has often remained limited to the domestic scale [19], City [20], public spaces [21], isolating institutional settings [22], psychological well-being [23], and therapeutic landscapes [24]. A comprehensive framework that bridges neurological findings with spatial features across various scales (from the home to the city) remains markedly absent.

In response to this gap, this article introduces the Neuro-Psycho-Contextual (NPC) Model. This meta-paradigmatic approach demonstrates how synergy between

neuroplasticity, procedural memory, behavioral habits, and contextual design helps compensate for episodic memory deficits and preserve individual agency. The ultimate goal is to present a model for Place-Based Policy that considers the environment as part of the "Life Space". From this viewpoint, "Aging in Place" is not merely remaining at a physical point, but ensuring the continuity of identity and the possibility of "Dwelling" in the world [25]. This paper integrates neurological, psychological, and contextual layers across different spatial scales through interdisciplinary synthesis, opening a new path for understanding cognitive aging with an emphasis on the latent capacities within space.

Theoretical Foundations and Methodological Position

The Convergence of Neuroscience, Ecological Psychology, and Socio-Spatial Structure

NPC is established upon the confluence of three complementary domains: the neuroscience of habit, ecological psychology (embodied perception), and socio-spatial theories. This integration aims to redefine the environment as a "neuro-cognitive supplement" that shifts the processing load from vulnerable systems of conscious cognition to the more resilient capacities of the body, habit, and environmental cues.

At the neurological level, this model relies on research concerning learning systems and habit formation. Studies by Graybiel [26] demonstrate that the repetition of action within a stable environment shifts behavioral control from the higher cerebral cortex to more resilient subcortical structures, specifically the basal ganglia. This automation creates a form of "cognitive energy economy", enabling the performance of daily activities with minimal conscious intervention. James [27] likens this mechanism to a "flywheel" that provides maximum stability for human action with minimum energy expenditure. In the context of dementia, this transfer of load to procedural memory is a vital pathway for preserving agency in the absence of episodic memory (Figure 1).

This neuro-foundational layer is phenomenologically and psychologically linked to the theory of embodied perception. Merleau-Ponty [28] argues that spatial perception is not an abstract mental representation but is deeply rooted in the "lived body" and "motricity". Accordingly, spatial cognition evolves through bodily interaction with the environment. Extending this perspective, Gibson’s theory of affordances [29] explains that the environment invites the body to act pre-reflectively through "action possibilities". Within this framework, environmental stability and legibility play a decisive role in reducing cognitive load and facilitating daily actions (Figure 1).

At the socio-ecological level, the interaction between body and environment is conceptualized through Bourdieu’s notion of "habitus" [30]. Habitus refers to sedimented behavioral and perceptual patterns stabilized within socio-spatial contexts, which exhibit resistance to neurodegeneration even in advanced stages of dementia [31]. From this vantage point, elderly independence is the product of a neuro-psycho-social ecosystem in which environment, body, and habit operate synergistically (Figure 1).

The synthesis of these three levels indicates that cognitive decline does not necessarily entail the collapse of agency, rather, it represents a shift in the mode of

action from conscious to embodied levels. Consequently, environmental design across various scales from the domestic to the urban can prevent the rupture between the individual and their life-space by providing familiar sensory cues and stable rhythms. Methodologically, this article constitutes a theoretical-interpretative study based on a critical analysis of interdisciplinary literature. The objective is to construct a conceptual bridge between clinical neuroscience, environmental psychology, and place-based health research to develop a translatable model applicable to environmental design, urban planning, and health policymaking.

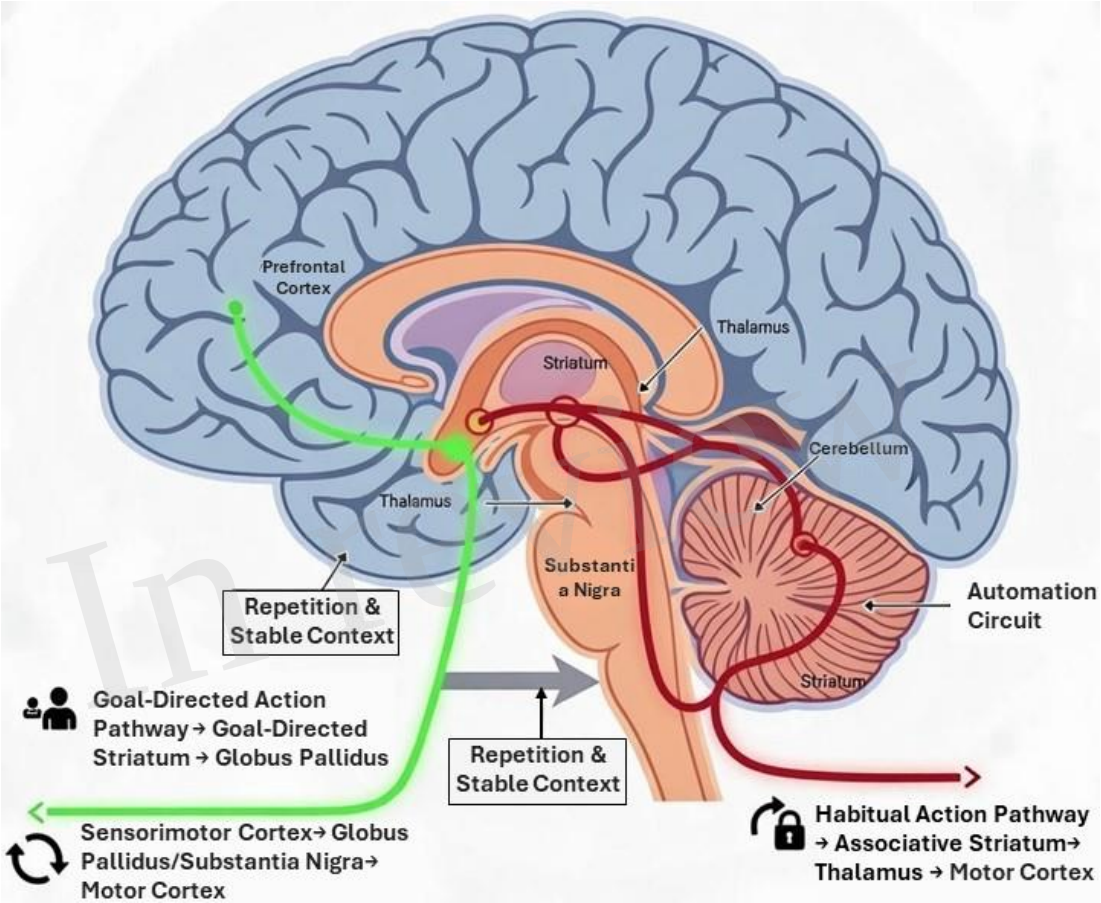


Figure 1. Neural Pathways of Habit Formation Based on Graybiel's Model

Interdisciplinary Conceptual Synthesis and Model Development

This research redefines cognitive aging as a dynamic and contextual phenomenon through an interdisciplinary conceptual synthesis. The focus of the study is not the accumulation of empirical evidence, but rather "conceptual engineering", a process aimed at constructing a coherent theoretical architecture that bridges abstract explanations with the practical levels of design and policy-making. This synthesis is based on identifying paradigmatic gaps and reorganizing them into an integrated model.

The starting point of this process is a critique of the dominant "patient-centered" and "brain-centric" approaches, which perceive dementia as an internal degradation disconnected from the life context (Figure 2). These approaches often neglect the role of environment, habit, and socio-spatial structures in maintaining or undermining cognitive function. In this re-reading, the core issue is redefined not merely as the

decline of cognitive capacities, but as the mismatch between an individual's residual capabilities and the spatial demands of the environment. In the subsequent stage, concepts extracted from neuroscience, ecological psychology, phenomenology, and socio-spatial theories were linked through a process of "layered synthesis." This synthesis explains how physical stimuli lead to automated neurological responses through embodied habits. Within this logic, "environmental plasticity" acts as a complement to neural plasticity, compensating for a portion of the limitations imposed by neurodegeneration. The result of this process is the preservation of embodied agency even under conditions of declining conscious cognition. In the final stage, these concepts were translated into an operational framework. NPC model is formulated to be functional across various scales from domestic housing to the urban fabric. As a conceptual scaffold, this model enables a rethinking of "cognitive justice" [32], as well as spatial and social justice [33][34], while strengthening cognitive resilience through the optimization of the everyday life-context (Figure 2).

In other words, this conceptual synthesis, through its translational logic, strengthens the connection between place-based health research, environmental design, and urban planning. It serves as a tool for moving beyond reductionist explanations toward a contextualized understanding of cognitive aging.

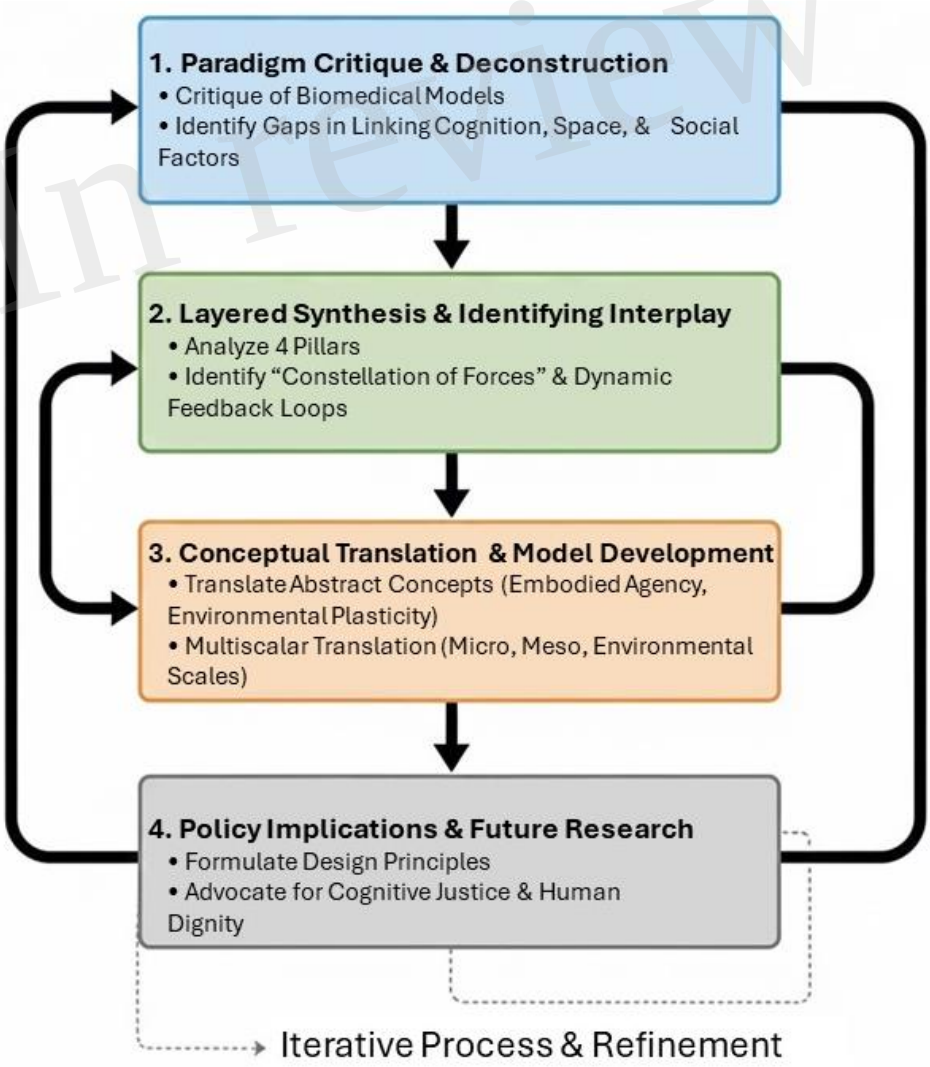


Figure 2. Methodological Stages: Interdisciplinary Synthesis for Cognitive Aging

Theoretical Grounding

The Neuroscience of Alzheimer's: Memory, Brain, and Plasticity

A precise understanding of Alzheimer's pathology, characterized by extensive synaptic degradation and the formation of amyloid plaques [35], is the primary prerequisite for redefining the concepts of aging and dementia [16]. Although episodic memory is severely damaged in the early stages, and semantic memory gradually deteriorates thereafter [7], procedural memory (responsible for practical skills and habitual behaviors) is remarkably preserved even into advanced stages [36]. This distinction in the vulnerability of memory systems provides the foundation for moving beyond a purely deficit-based approach.

On the other hand, the aging brain, even after the onset of dementia, possesses a form of inherent plasticity that allows for synaptic reorganization in response to environmental stimuli [16][37]. Evidence suggests that non-pharmacological interventions, including physical activity, cognitive routines, and enriched environments, modulate the trajectory of decline by activating this residual capacity [38]. This adaptive capability increases responsiveness to non-pharmacological treatments [39][35] and proves that even simple daily habits can maintain brain function in a "sharp" state [40].

The central premise of this research is that even in the face of fundamental structural changes, what Malabou terms "explosive plasticity" [14], habit-based design can sustain viability and establish a stable balance between permanence and change [15]. This approach aligns with Bergson's view [10], which suggests that the collapse of representational memory does not equate to the dissolution of ontological existence; rather, existence persists through bodily memory and the flow of action within the environment. Within the framework of embodied cognition, cognition is the result of continuous interaction between the body and the spatial environment [41]. From this perspective, spatial cues and sensorimotor stimulations can activate bodily memory as a "functional scaffold," ensuring locomotor independence and daily decision-making in the absence of episodic memory (Figure 3).

Behavioral Psychology and Habit: The Role of Habits, Cues, and the Environment

Preserving independence in aging extends beyond biological capacities, taking shape within the complex interaction between brain, behavior, and environment. A key mechanism in this interplay is the "habit loop," which significantly reduces the cognitive load of decision-making by automating daily routines [42]. According to James's theory [43], transferring the elements of daily life to the level of automation liberates cognitive faculties for more critical tasks. These loops, comprising environmental cues, behavioral routines, and neural coordinations [44], allow the environment to guide the process of action pre-reflectively through context-triggered behaviors. In this sense, habit is not merely repetition but a form of low-cost, sustainable action [45].

Neurological findings indicate that this process involves a shift in behavioral control from associative loops to sensorimotor loops within the basal ganglia [42]; structures capable of reproducing learned behaviors without conscious intervention [26] (Figure 1). This structural dissociation explains why habits are preserved even when the

higher cerebral cortex undergoes degradation. Furthermore, conscious focus on automated skills can lead to performance decrements [42], underscoring the necessity of designing environments that elicit action through sensory affordances rather than conscious processing.

Concurrently, the intrinsic motivation of the elderly is bolstered by the maintenance of independence [8]. Environments that are simultaneously predictable and appropriately challenging, relative to an individual's capacity, strengthen the sense of autonomy and reduce stress [46]. Beyond habit, social interactions and emotional responses to the environment play a vital role in cognitive health; group activities strengthen working memory [47], and interaction with natural spaces improves quality of life [48][49].

From a psycho-social perspective, this process aligns with Fromm's view on the human need for a "frame of orientation and devotion" [50]. Habit-based environments provide a sense of security by reconstructing this framework which, according to Maslow's hierarchy, serves as the foundation for achieving self-esteem and agency [51]. Thus, by acting as an intermediary between neural structures and the spatial context, this layer transforms independence from a conscious effort into an embodied and internalized state (Figure 3).

Contextual Layer: The Influence of Social, Cultural, and Spatial Environments on Cognition

Recent studies have demonstrated that environmental characteristics influence brain structure and function through neurobiological pathways, such as neurotrophic signaling and the attenuation of oxidative processes [12]. Within this framework, habit-based design emerges as a key trajectory for achieving positive brain aging. Here, the environment is perceived not as a static backdrop, but as an active agent in cognitive health [20] that simultaneously encompasses social, physical, and sensory dimensions.

The socio-spatial dimension of this layer includes elements such as the stability of cues, environmental legibility, and the controlled diversity of sensory stimulations. Fixed navigational cues shift the cognitive load of orientation to procedural memory, transforming wayfinding from a conscious decision into a sensorimotor sequence [49]. From this perspective, the role of the environment extends beyond local problem-solving in navigation [30]; spatial agency is the product of the interaction between environmental structure, prior experiences, and an individual's affective capacities. According to the ecological-enactive approach [29], the environment functions as a "cognitive scaffold," creating a foundation upon which the mind can lean [16]. At a philosophical level, Heidegger [52] explains how familiar environments transform space from an alienated state into a "home" and a place of "dwelling". Similarly, Casey [53], by introducing the concept of "place memory," emphasizes that places carry memories, and spatial cues at the neighborhood and urban scales serve not only as guides but as preservers of identity.

The socio-cultural dimension of this layer addresses the issue of spatial inequities. Inclusive and participatory public spaces can reduce place-based disparities in health and assist in the stabilization of daily routines [49]. However, environmental design must be calibrated to avoid both over-stimulation (leading to agitation) and under-stimulation (leading to social isolation) [22]. Ultimately, by expanding the

concept of "life space" and planning for safe mobility at the local and urban scales, the physical environment can become an accelerating factor that, in a dynamic cycle, reinforces both neuroplasticity and psychological stability (Figure 3).

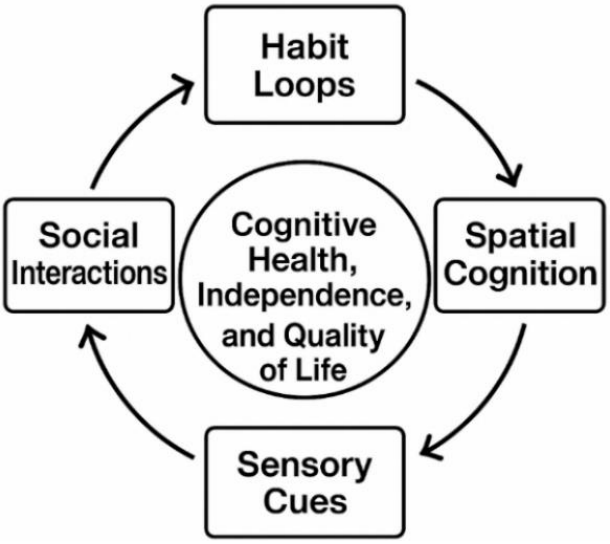


Figure 3. Contextual Pathways to Cognitive Health

In review

NPC Conceptual Model

From Neural Structure to Spatial Agency

As previously established, the NPC model is predicated on the assumption that promoting cognitive health in aging is not a single-layered intervention. Instead, it is the outcome of a dialectical interaction between three layers: neurological, psychological, and contextual (socio-environmental).

In the neurological layer, the model utilizes functional memory dissociation to shift the processing load from damaged pathways to preserved mechanisms. At this level, habit functions as a neural shortcut that, by leveraging neuroplasticity, enables functional continuity through "learning without remembering." Consequently, even as episodic memory declines, functional capabilities remain stable through procedural circuits. Simultaneously, in the psychological layer, the environment transforms from a neutral backdrop into an active agent in behavioral regulation. By organizing cues, space triggers habit and reduces the need for conscious, anxiety-inducing decision-making. This process, in addition to ensuring "attention economy," stabilizes the individual's bond with the environment by reinforcing a sense of self-efficacy [54]. Finally, the contextual layer acts as a cognitive-motor scaffold by providing cue stability and spatial clarity, facilitating independent navigation across various scales by activating embodied memory.

The primary innovation of this model lies in moving beyond the isolated analysis of these layers, focusing instead on the "Interplay" as its operational core. In this interpretation, personality stability and functional independence are achieved through the synergy of three mechanisms (Figure 4):

Neuro–Contextual Scaffolding: Relying on embodied cognition, the physical environment becomes a form of external memory. Stable spatial cues shift navigational processes from conscious decision-making to automated perceptual-

motor sequences. In this way, independent activity remains possible even in the presence of hippocampal atrophy.

Psycho–Contextual Habituation: This mechanism explains the link between the socio-spatial layer and behavioral structures. A predictable and meaningful environment elicits learned routine behaviors through simple, familiar stimuli. By strengthening the sense of self-efficacy, it integrates space into the individual's lived experience and identity, an identity that serves as a protective shield even against memory loss.

Neuro–Psychological Synergy: At this level, the mutual regulation of neurological states and psychological responses occurs through environmental stimuli. Controlled sensory stimulations such as rhythmic lighting or access to green space act through environment-driven plasticity to influence reward and motivation circuits. Consequently, active participation increases, and the severity of Behavioral and Psychological Symptoms of Dementia (BPSD) decreases.

At the convergence of these three pathways, the core of "Functional Independence" is formed. When an environmental scaffold aligns with established habits, neurological deficits do not necessarily lead to functional collapse. From this perspective, the environment is not inherently therapeutic unless it operates in active conjunction with psychological and neural layers [54]. This logic transforms the model into an operational tool for the design, policymaking, and planning of environments for the aging population.

In review

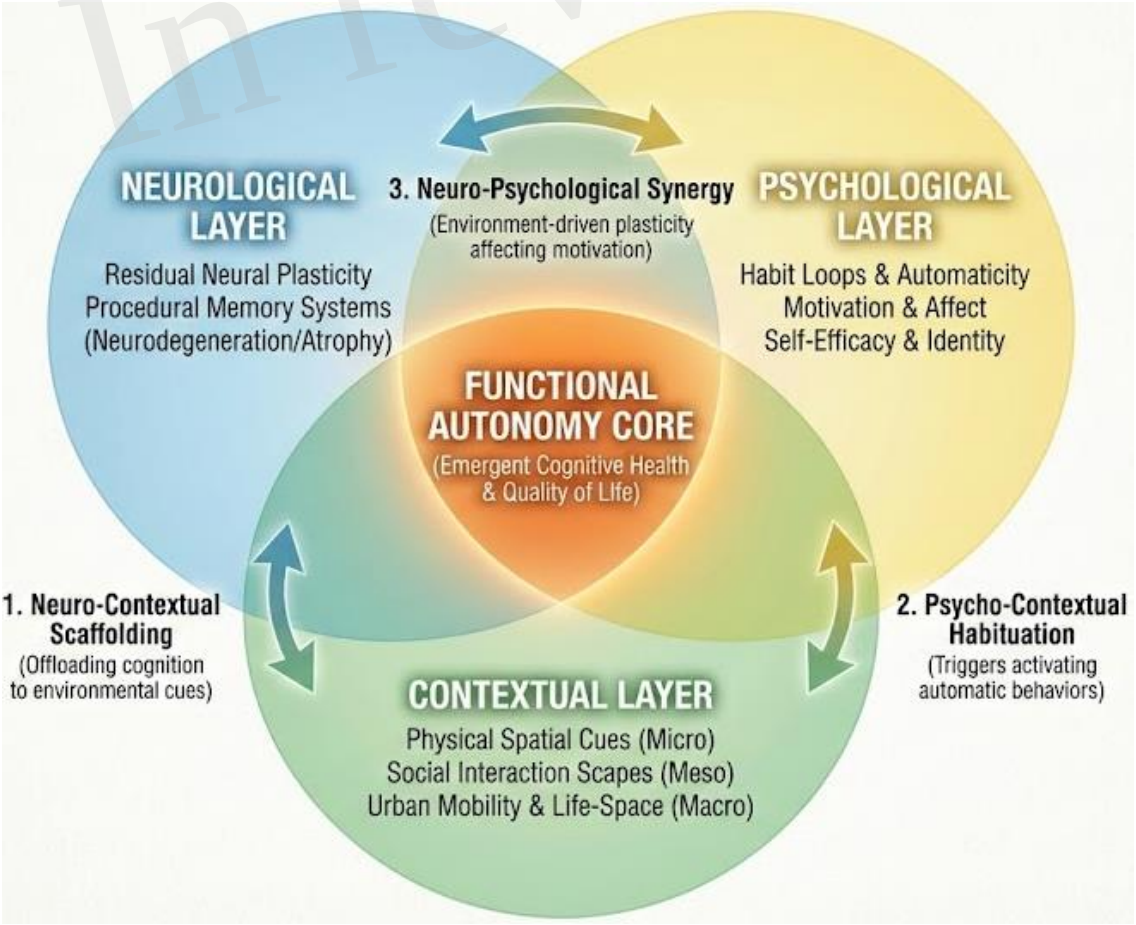


Figure 4. NPC Conceptual Model

Translational Principles for Model Application

To operationalize NPC model across the scales of the home, neighborhood, and city, adherence to three complementary principles is essential. These principles elevate the link between cognitive health and place from a theoretical level to an interventional strategy across various spatial scales.

Empowerment-based Design: This principle emphasizes focusing on an individual's preserved capacities rather than merely their deficits. The environment must be configured to activate residual skills, transforming the individual from a "passive recipient of care" into an "active agent" in daily life. In this framework, the environment functions as a scaffold that, as Vygotsky [55] suggests, possesses the capacity to support higher mental functions.

Habit-strengthening Environments: In this principle, neuroscientific knowledge regarding the formation and persistence of habits is directly translated into spatial design. Intuitive spaces, familiar cues, and stable layouts enable the individual to rely on procedural memory and a sense of place, thereby reducing dependence on volatile episodic memory. Consequently, functional independence is reinforced at the domestic, neighborhood, and urban scales; as James posits, habit yields maximum efficiency with minimum cognitive energy [43].

Person-centered Personalization: This principle emphasizes reflecting identity, lived history, and individual preferences within the environment. As episodic memory declines, maintaining the continuity of the "Self" (Ipse) [9] becomes vital for the dignity of the elderly. An environment that carries familiar and meaningful cues serves not only as a guide for spatial action but also as a preserver of the continuity of identity; for, as Casey [53] explains, places are the bearers of memory and meaning.

These three principles form a dynamic framework that places the interaction between the brain, mind, and environment at the heart of cognitive health. The result of this approach is the provision of practical solutions for the design of care environments and urban policymaking aligned with WHO standards [17][18], an approach that transforms every environmental intervention into a step toward achieving "Cognitive Justice" in aging.

Architectural Implications: Spatial Engineering as a Technology of Agency Regulation

NPC model redefines architecture from a mere physical intervention into a "technology of agency regulation". Within this interpretation, cognitive decline is not solely a neurobiological trajectory but a consequence of the imbalance between brain capacities and spatial matrices. Architecture is not a passive backdrop, but it is part of a "power-cognition ecology" that defines the boundaries of the subject's ability to be and to act. From this perspective, spatial design serves as a tool for reconfiguring the alignment between the elderly subject and the environment across the life course, determining who can act in the world, how, and to what extent. Consequently, spatial design is elevated to a level of intervention in the production of the "elderly subject," where architecture acts as an active mediator in regulating agency throughout the aging process.

Dwelling as a Distributed Cognitive Infrastructure

This research reframes the home not merely as a physical container, but as a distributed cognitive infrastructure at the micro-scale. Space functions as an apparatus for the production of habit, embodied memory, and the regulation of daily action. Spatial organization, motor sequences, and sensory cues generate "spatial scripts" that guide the body without relying on conscious representation, thereby scaffolding cognition outside the brain.

In this reading, the home is part of a distributed cognitive architecture that shifts the processing load from vulnerable episodic memory to procedural and embodied mechanisms. This approach is grounded in the logic of Person–Environment Fit, transforming the house from a passive shelter into an empowering environment. Spatial stability, functional legibility, and environmental predictability allow for the formation of automated perceptual-motor sequences, such that daily behaviors are organized without the need for conscious recollection.

From this viewpoint, the stable placement of objects, the reduction of visual clutter, and guiding sensory cues are not aesthetic decisions but essential components of the infrastructure for preserving agency. Functional nodes within the home such as nutrition, personal hygiene, and medication management are conceptualized as critical points in the cognitive economy of daily life, reproducing the boundary between independence and dependence. Disruptions in Activities of Daily Living (ADLs) are often the primary indicators of declining cognitive independence [56]. Therefore, the organization of these nodes is elevated to a level of micro-cognitive policy, determining whether the elderly person remains an active agent or is transformed into a "subject of care". Spatial interventions at this level must enable active coping rather than replacing the individual's function. This implies a space where the individual remains an actor, albeit with minimal cognitive strain [19]. In this sense, through affordances [29], the home becomes an extension of the cognitive system rather than a substitute for it.

The Neighborhood as an Infrastructure of Social Memory and the Architecture of Motor Identity

At the neighborhood scale, space functions as an active external memory and a distributed socio-cognitive infrastructure [57] that simultaneously supports navigation, mobility, and the continuity of identity. NPC model perceives the neighborhood not merely as a social backdrop, but as a distributed cognitive system. Through spatial cues, rhythmic patterns, and repetitive movement sequences, it facilitates the continuity of the elderly person’s subjecthood and agency.

Daily movement within the neighborhood is not merely physical displacement, but it is a form of socio-cognitive exercise that embeds the body within a network of meaning, belonging, and memory. Within this framework, cognitive disorientation is regarded not simply as an individual deficit, but as a rupture in the place-identity bond. Therefore, neighborhood design acts as a technology for reconstructing the link between body, memory, and community, fostering the conditions for active and participatory aging beyond the domestic sphere [47].

Environmental legibility and predictable structures facilitate automated motor patterns and mitigate navigational anxiety, transforming the security of movement

into a habitual quality [58]. Space serves not only as a directional guide but as a bearer of meaning and memory, contributing to the reproduction of spatial identity and social belonging [59].

Design strategies include distinct spatial landmarks at decision nodes, "recharge stations" acting as cognitive reset points, and legible, looped pedestrian paths that reduce the fear of getting lost. These elements transform independent mobility into the habitual flow of active aging. Visual, olfactory, and auditory cues function as identity affordances, simultaneously stabilizing social interaction and spatial attachment [29]. Thus, the neighborhood evolves into a dynamic network of embodied memories, integrating cognition, movement, and society into a unified spatial ecology. In this reading, urban design at the meso-scale constitutes a form of cognitive policy that redefines the boundaries of independence, belonging, and social participation for the elderly.

City as an Ecology of Plasticity and an Infrastructure of Cognitive Justice

At the urban scale, the model manifests as an ecology of plasticity that emphasizes the empowerment and sustained presence of the elderly in public spaces, rather than their control or segregation. The objective is to realize Aging-in-place within a context of spatial and social diversity, mitigating risks without eliminating lived experience.

Predictable spatio-temporal rhythms, legible infrastructures, continuous green networks, and "Forgiving Environments" transform cognitive errors into manageable situations rather than triggers for spatial exclusion [58]. By regulating biological rhythms, reducing anxiety, and stabilizing habitual motor patterns, these infrastructures enable an embodied presence in public spaces. The urban light-space rhythms can function as an environmental metronome, regulating circadian cycles to reduce spatio-temporal cognitive disruptions [60]. Multisensory nature corridors serve as infrastructures for environment-driven plasticity, modulating anxiety and BPSD while exposing the elderly to unplanned social interactions [61]. Legible and embodied transportation systems utilizing stable color-coded and auditory cues, facilitate habitual orientation and reduce reliance on complex cognitive processing [29]. Forgiving environments, equipped with multi-layered decision nodes, directional tactile paving, and local affective landmarks, allow for error recovery without stigmatizing surveillance.

Consequently, within this model, the city is not merely a care-giving backdrop but an apparatus for the production of agency. It can function either as a machine for cognitive exclusion or a machine for social participation. Urban design is not an attempt to "eliminate" dementia, but rather to reconstruct agency amidst cognitive instability [11].

Artificial Intelligence as a Secondary Cognitive Layer

The implementation of multi-scalar interventions necessitates the utilization of Artificial Intelligence as an analytical and operational infrastructure. Machine learning algorithms can extract embodied motor patterns, life-space trajectories, and micro-behavioral changes across the scales of the home, neighborhood, and city, enabling the design of adaptive and personalized environments.

At the domestic scale, intelligent systems can dynamically calibrate spatial affordances based on an individual's habitual patterns. At the neighborhood scale, the analysis of mobility and social data allows for the reconfiguration of path networks and cognitive nodes. At the urban scale, predictive models identify place-based cognitive inequities and support justice-oriented spatial policies. Consequently, AI functions not as a surveillance technology, but as a secondary cognitive layer that reinforces the co-evolution of the brain, body, and environment within a data-driven cognitive ecology. Notably, this subject could serve as an independent interdisciplinary research area, where the role of AI in establishing design protocols for dementia care can be explored more profoundly.

Policy Implications and Future Outlook: Redefining the Spatial Governance of Cognitive Health

NPC model facilitates a rethinking of the spatial epidemiology of aging, introducing the "Context × Cognition" interaction as a structural determinant in aging trajectories. This necessitates a paradigmatic shift in health policies: moving from the pharmacological management of symptoms toward the organization of empowering environments.

Spatial governance must reconfigure residential and urban environments as legible cognitive infrastructures that compensate for episodic memory vulnerabilities by reinforcing embodied memory [17][18]. It must be noted that the mere idealization of the "home" concept, without considering social inequities, can exclude vulnerable groups from the path of spatial justice [19]. Therefore, policymaking must transition from providing passive care services toward place-based active empowerment. Empowering environments can delay institutionalization and significantly enhance the quality of life [31]. To this end, housing and health authorities should rewrite construction regulations based on neuro-environmental models, treating spatial design as a formal part of the "clinical prescription."

This study introduces the emergent concept of "Spatial Cognitive Justice". This term refers to the fundamental right of all individuals, including those living with dementia, to perceive, comprehend, and manage their living environments. The concept demonstrates that inequities in spatial perception are rooted in the space itself. Unequal access to empowering environments creates divergent trajectories of cognitive decline or stability. Thus, space becomes a structural variable in the reproduction of health disparities [62][63].

The distinction between this approach and traditional discourses on spatial justice [64] lies in its focus. While traditional spatial justice emphasizes equal access to resources and services [33][34] or the right to knowledge production [32], Spatial Cognitive Justice emphasizes the subject's capacity to understand and manage their own environment. Consequently, environmental design and urban planning are elevated to the level of cognitive policymaking, tasked with the structural redefinition of justice and the provision of a more equitable and empowering experience of the world for the elderly.

Discussion

This study has developed an integrated model for understanding the interactions between neural infrastructures, psychological processes, and environmental

contexts throughout the trajectory of cognitive aging. NPC model represents a paradigmatic shift from reductionist biomedical approaches toward place-based and embodied ontologies of aging. Functional decline emerges not merely as brain pathology, but as a complex interaction between neurological, psychological, and environmental layers. This shift in perspective is vital because, despite advances in neuroscience and pharmacological interventions, the growing dementia crisis remains without a comprehensive response, and conventional "Aging in Place" policies often limit the issue to survival and safety [19][20].

This is particularly critical in societies where, due to development and progress, life expectancy has risen, fundamentally intertwining the concept of human society with the concept of aging. Consequently, alongside general concepts such as gender, aging will emerge as a primary analytical category. The findings of this study align with Sabat [6], who argues that the term memory loss is a misleading description of dementia that can lead to social and caregiving misconceptions. This research demonstrates that individuals with dementia, despite impairments in explicit memory, preserve certain cognitive strengths and implicit memory [7][8]. Thus, the proposed model redefines dementia and cognitive aging as a process amenable to environmental scaffolding; that is, the design of empowering environments can activate residual cognitive abilities and enhance lived experience. By leveraging this capacity, the model proposes environments organized to maintain habit loops, spatial cues, and daily routines.

Furthermore, by rejecting behaviorist views that perceive the environment as a stimulus and individuals as reactive agents, this study conceptualizes the environment as opportunity structures (affordances) that activate embodied agency [29]. When episodic memory impairment exacerbates feelings of helplessness, environments designed around bodily memory restore the "I can" experience, supporting basic self-awareness and existential continuity. Moreover, the model builds upon classical traditions such as Lewin's Field Theory [65] ($B = f(P, E)$), demonstrating how environmental forces shape behavioral choice. This framework extends the foundational traditions linking the self, space, and memory, showing how design can recalibrate these relationships toward dignity and agency.

Existing literature, such as studies by Aslanoğlu [20] and Hutson & Hutson [16], has primarily focused on macro-environmental factors such as density, service accessibility, and pollution. Meanwhile, micro-scale layers such as homes, daily pathways, semi-private spaces, and sensory cues have remained understudied. The NPC model addresses this gap by introducing the "Interplay" as a multi-scalar analytical lens that integrates housing, care spaces, social networks, and life-space dynamics.

Additionally, this model confirms the findings of Ward et al. [11], asserting that the environment must maintain the capacity to reconstruct the elderly person's relationship with place even after cognitive impairment, by promoting bodily autonomy and connection with the environment's affective identity. Consequently, Non-Pharmacological Interventions, such as environmental design and multisensory stimulation, are positioned not as auxiliary tools but as primary care mechanisms that target preserved cognitive pathways and provide alternative routes for empowerment. This research establishes a common conceptual language between architecture, neuroscience, and policymaking, enabling the integration of smart

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technologies [66], multisensory interventions [16], and pharmacological approaches [15] into a unified cognitive ecology.

While empirical validation requires mixed-methods approaches including geospatial analysis, embodied mobility tracking, and longitudinal life-space studies, this framework provides a systematic analytical infrastructure for assessing brain–behavior–environment interactions and place-based cognitive inequities.

Conclusion, Limitations, and Future Research Directions

By formulating the NPC model, this research has demonstrated that functional independence in elderly individuals with Alzheimer's is less a function of biological degradation and more a product of the interaction between residual neural capacities and environmental scaffolding. This model proves that architecture and urban planning can function as an active therapeutic intervention, shifting the cognitive load from damaged brain regions to the environment. The primary conclusion is that through the routinization of space and the reinforcement of habit loops, self-efficacy can be preserved even in the presence of hippocampal atrophy, transforming the concept of Aging in Place from a mere ideal [19] into a clinical-spatial reality.

The fundamental distinction of this study from previous approaches lies in its focus: while current smart technologies primarily concentrate on "detecting" changes in habits for reporting purposes [66], the proposed model emphasizes "guiding" and "supporting" habits through spatial organization. Consequently, habit is transformed from a simple behavioral output into a neurological survival strategy, where the physical environment and digital systems function as a Hybrid Ecology to compensate for memory deficits.

Despite its theoretical contributions, this research faces several limitations. First, the interdisciplinary nature of the model presents logistical and ethical challenges in the simultaneous collection of neurological data (such as fMRI) and macro-spatial variables (such as GIS). Second, individual differences in Cognitive Reserve and cultural backgrounds may influence the rate of new habit formation, which the current model assumes to be uniform. Furthermore, adapting environmental protocols to diverse cultural and social contexts requires more extensive investigation.

Accordingly, future research should focus on the empirical testing of the link between environmental features and brain plasticity. The use of Virtual Reality (VR) to simulate design interventions and assess the behavioral responses of the elderly prior to physical implementation could be groundbreaking. Additionally, investigating the role of AI-based assistive technologies in strengthening the Psycho-Contextual layer and providing continuous Life-space monitoring opens new horizons for the personalization of environmental interventions within the framework of Spatial Cognitive Justice.

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